

Statistical Applications

Store Choice and Customer Behaviour: Multinomial Logit Analysis

By Anton Cronet

Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
5. Implications and Limitations

1. Introduction - Background

Fast Fashion:

Low prices

Fast-changing trends

Mass production

→ Accounts for ~10% of global CO₂ emissions (Niinimäki et al. 2020)

Attitude - Behavior Gap:

Consumers claim to value sustainability

Behavior remains price-driven (Carrington et al. 2010)

Research Question:

→ What is most important to consumers?

1. Introduction - Theoretical Background

Random Utility Theory:

Choice is based on utility maximisation

Choices will have a random element to them

Utility = systematic component V
+ random error ε

Discrete Choice Experiments:

Modelling Choice was introduced by McFadden (1974)

Based on Random Utility Theory

Assumes errors are Gumbel distributed

→ Allows for choice to be analytically modelled

Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
5. Implications and Limitations

2. Data Description - Summary Statistics

Discrete Choice Experiment
Survey - based

Respondents are MLU students

Choose a **Black T-Shirt** between:
Store 1, Store 2, or neither

Only those who answer fully are kept:

- 319 individuals
- 6 discrete choices each
- 1,914 total observations

Attribute	Store 1	Store 2
Price	€39.95	€19.95
Headquarter Location	USA	EU
Distribution Channel	In-Store	Online
Shipping Time	0 Days	1 Day
Marketing Strategy	Moderate	High
Ecological Sustainability	✗	✓
Social Sustainability	✓	✗

2. Data Description - Variables

Individual information:

Id, Gender, Age,

Household Size and Income,

Monthly Clothing Spending and
Amount, of which in-store?

Second hand purchases (in
general and/or for events)

Store Information:

Price, Shipping Time (days),

Sustainability (Social / Ecological),

Headquarters
(Germany / EU / US / Asia),

Distribution
(Online / In-store / Hybrid),

Marketing (Min / Mod / Max)

Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
5. Implications and Limitations

3. Methodology - Assumptions

Why is MNL Appropriate?:

- Discrete choice structure
- Independence of Irrelevant Alternatives (IIA) assumption
- Random Utility Theory assumption
- Panel data structure

IIA is confirmed via Hausman-McFadden test:

Reestimate model - remove alt 3

χ^2 test to see if Betas are the same

Chi-squared: 3.0014

Degrees of freedom: 1

p-value: 0.0832

Result: Fail to reject H0 ☐ **IIA Holds**

3. Methodology - Utility Function

Utility Function: $U_{ij} = V_{ij} + \epsilon_{ij}$

Systematic component: $V_{ij} = ASC_j + \beta_{indiv} * Individual_i + \beta_{store} * Attributes_{ij}$

U : total utility consumer i gets from store j

ASC = Alternative Specific Constant : baseline preference for store j over Store 3 (where $U = 0$)

ϵ : random component, assumed i.i.d. Gumbel distributed $\square$ produces MNL closed-form probabilities

Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
5. Implications and Limitations

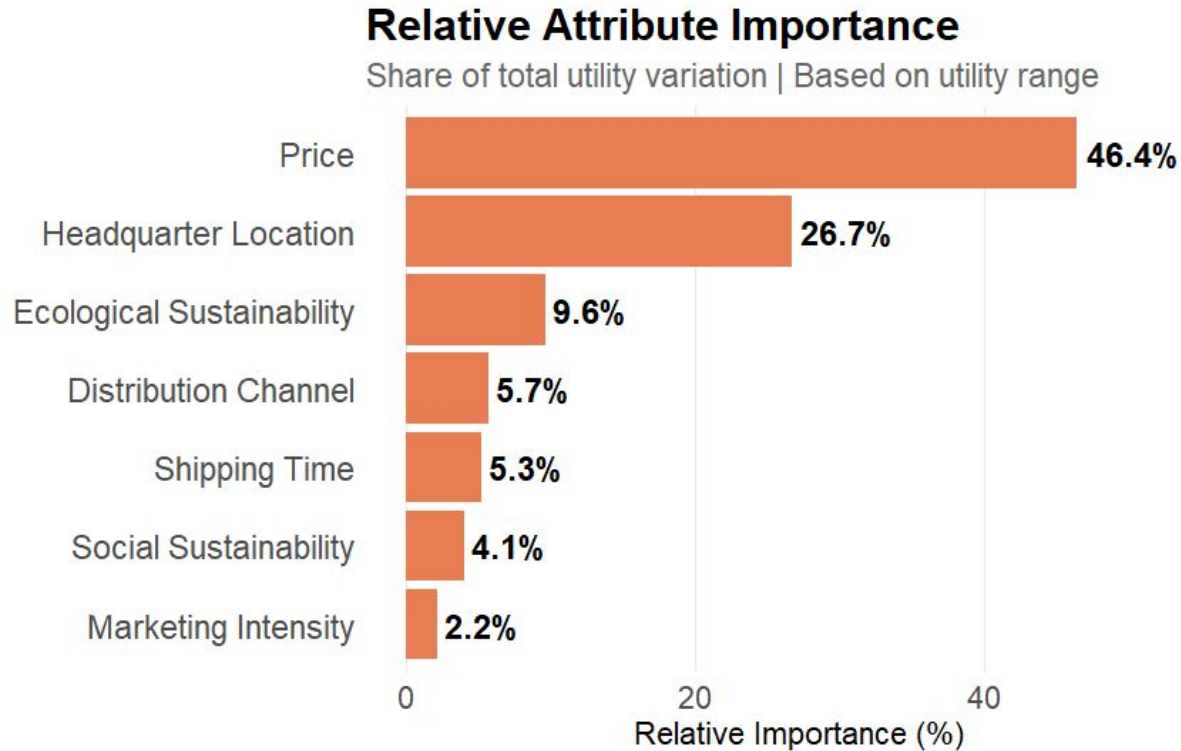
4. Results - Store Estimates

Variable	Estimate	Robust s.e.	Sig.
Price (€)	-0.052	0.005	***
Ecological Sustainability	+0.396	0.059	***
Social Sustainability	+0.167	0.057	***
HQ: Asia (ref: Germany)	-1.095	0.111	***
HQ: US (ref: Germany)	-0.667	0.113	***
HQ: EU (ref: Germany)	-0.609	0.105	***
Distribution: Online (ref: In-Store)	-0.223	0.098	**
Distribution: Hybrid (ref: In-Store)	+0.013	0.069	
Shipping Time (per day)	-0.027	0.012	**
Marketing: Maximum (ref: Minimal)	-0.091	0.075	
Marketing: Moderate (ref: Minimal)	-0.037	0.079	
Significance: *** p<0.01, ** p<0.05			

Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
 - a. Relative Utility Range
 - b. Willingness To Pay
 - c. Store Price Elasticity
 - d. Individual Variables
5. Implications and Limitations

4. Results - Attribute Importance



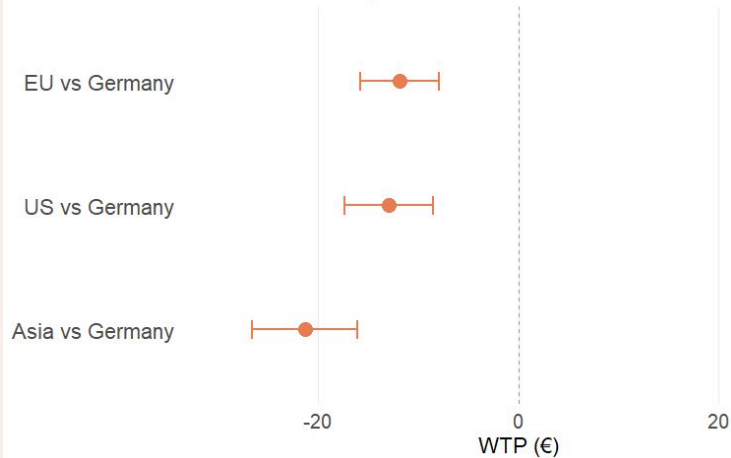
Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
 - a. Relative Utility Range
 - b. Willingness To Pay
 - c. Store Price Elasticity
 - d. Individual Variables
5. Implications and Limitations

4. Results - Willingness To Pay

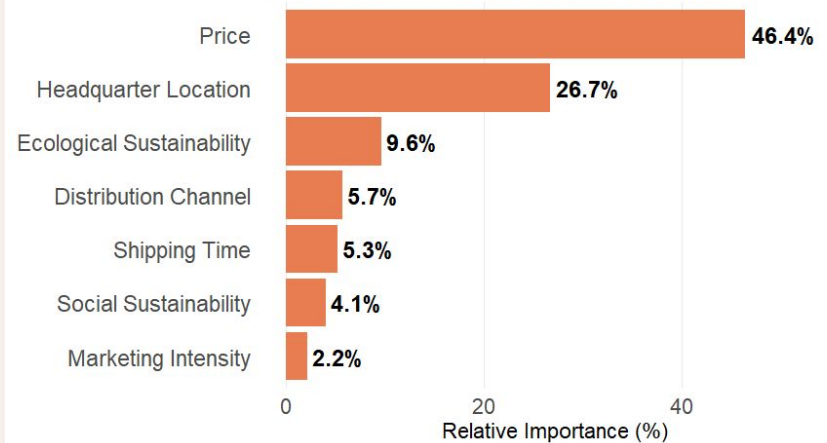
WTP – Headquarter Location

Reference: Germany



Relative Attribute Importance

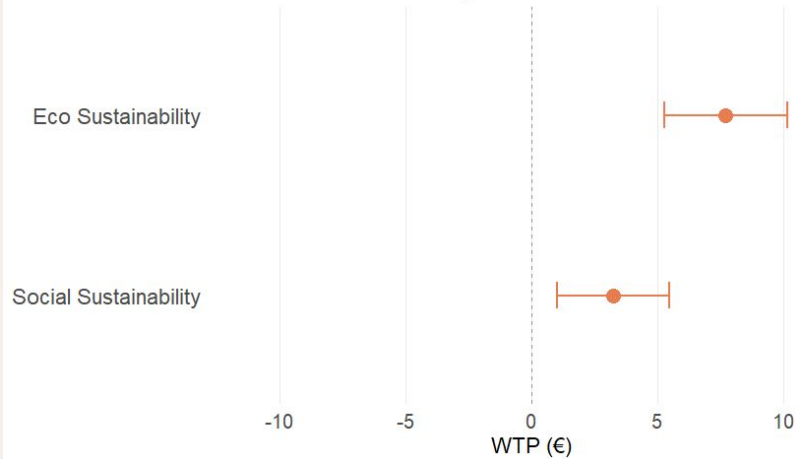
Share of total utility variation | Based on utility range



4. Results - Willingness To Pay

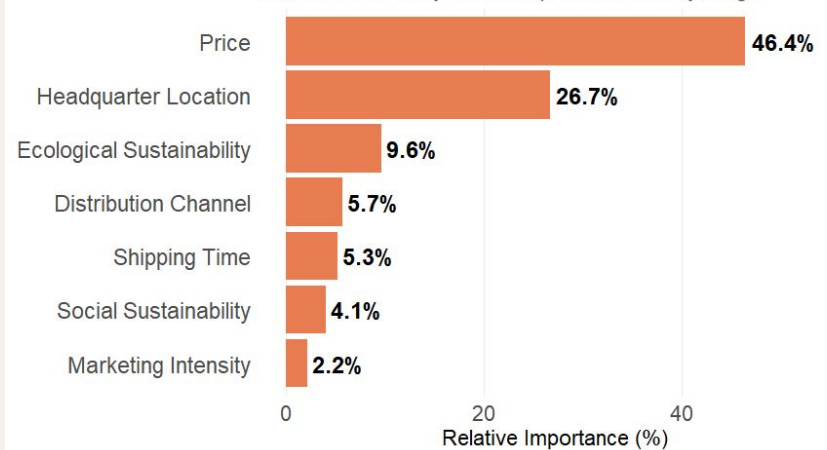
WTP – Sustainability

Reference: No sustainability commitment



Relative Attribute Importance

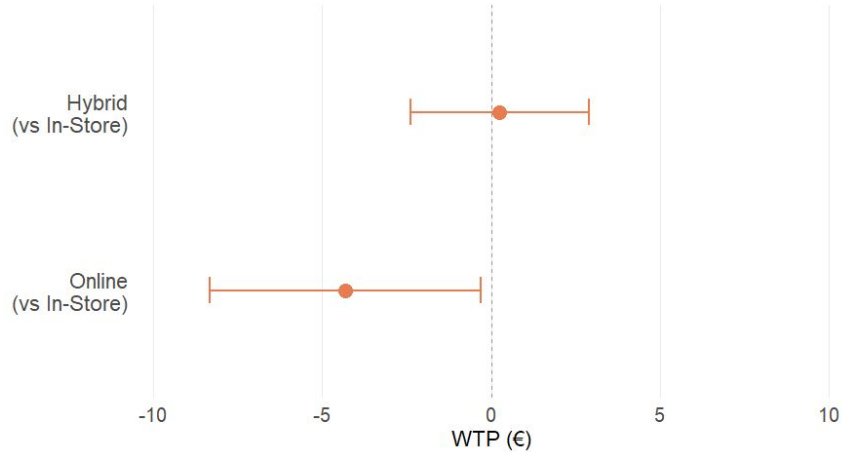
Share of total utility variation | Based on utility range



4. Results - Willingness To Pay

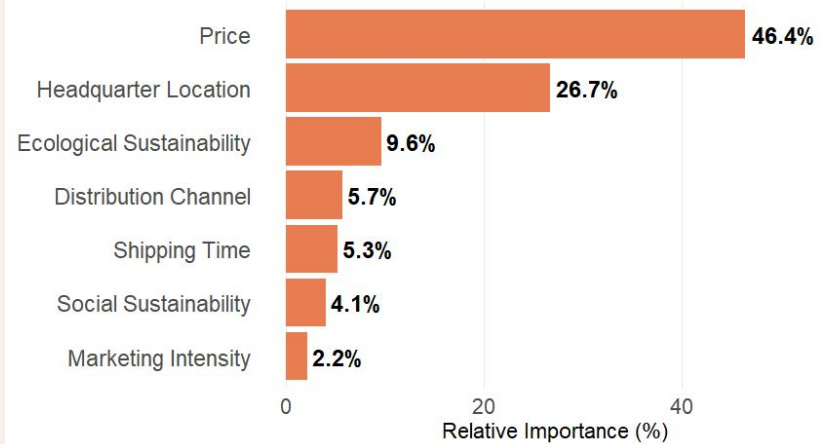
WTP – Distribution Channel

Reference: In-Store

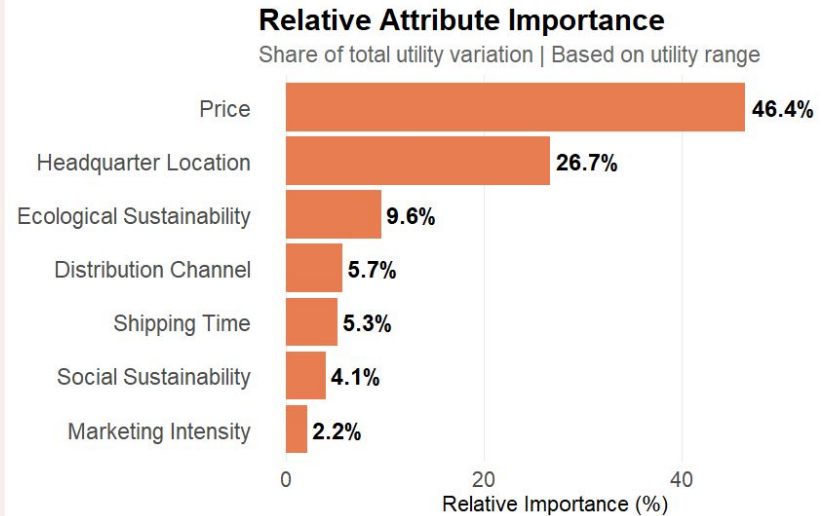
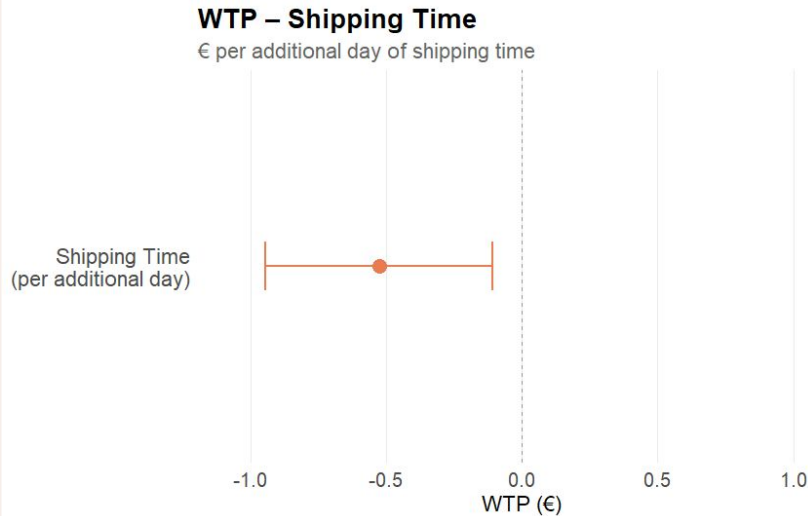


Relative Attribute Importance

Share of total utility variation | Based on utility range



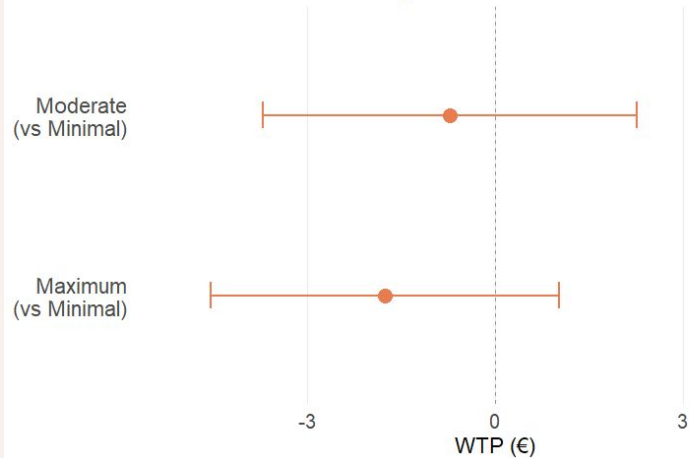
4. Results - Willingness To Pay



4. Results - Willingness To Pay

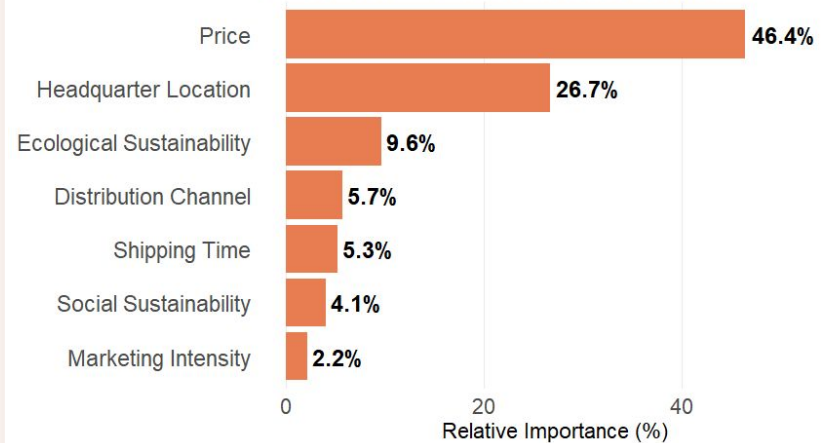
WTP – Marketing Intensity

Reference: Minimal marketing



Relative Attribute Importance

Share of total utility variation | Based on utility range

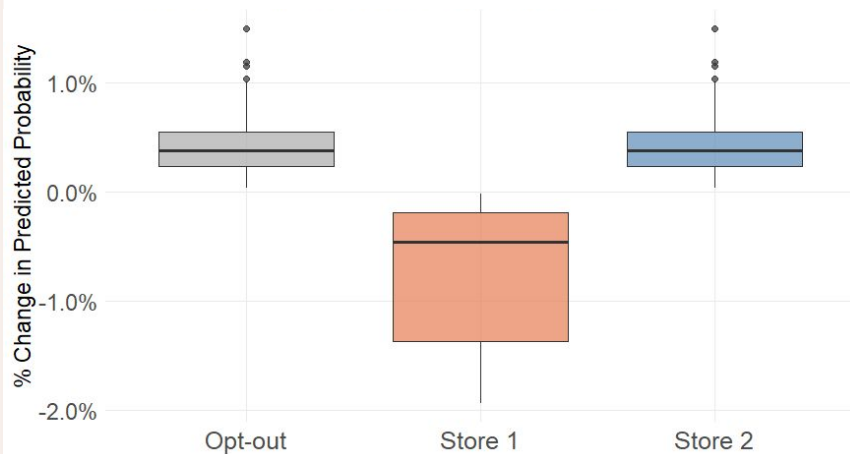


Outline

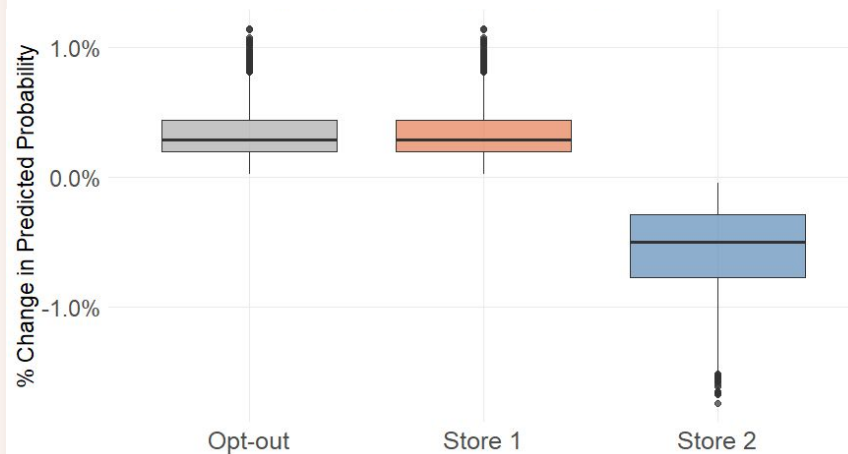
1. Introduction
2. Data Description
3. Methodology
4. Results
 - a. Relative Utility Range
 - b. Willingness To Pay
 - c. Store Price Elasticity
 - d. Individual Variables
5. Implications and Limitations

4. Results - Store Price Elasticity

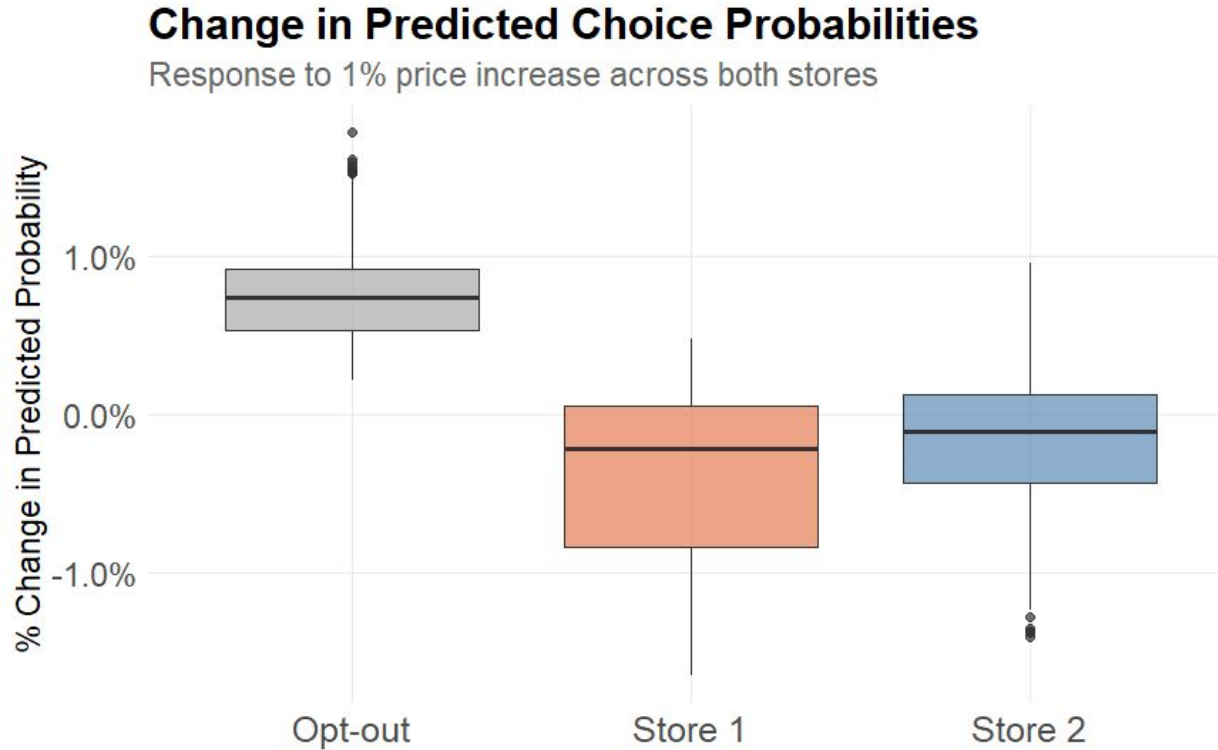
Store 1: 1% Price Increase



Store 2: 1% Price Increase



4. Results - Store Price Elasticity



Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
 - a. Relative Utility Range
 - b. Willingness To Pay
 - c. Store Price Elasticity
 - d. Individual Variables
5. Implications and Limitations

4. Results - Individual Estimates

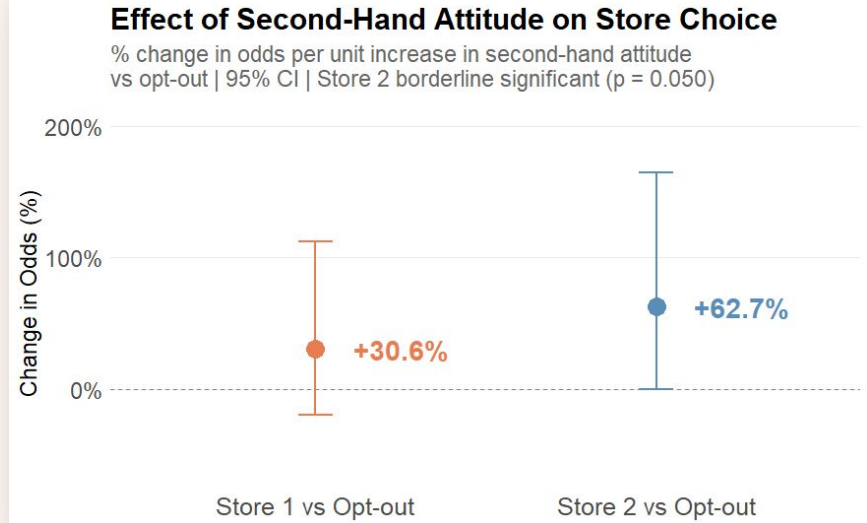
Variable	Estimate	Sig.
Age	-0.031	***
General Clothing Spend	+0.009	***
Second-Hand (For events) - Store 2	+0.486	**
In-Store Shopping Share	-0.014	*
All others	n.s.	
Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$,		

4. Results - Individual Characteristics

Individual characteristic importance can not be assessed like store characteristics

Logit model is not directly intuitive

→ Interpret as % change in odds



4. Results - Individual Characteristics

Effect of Age on Store Choice

% change in odds per additional year of age
vs opt-out | 95% CI



Effect of Monthly Clothing Spend on Store Choice

% change in odds per unit increase in monthly
stationary retail spend vs opt-out | 95% CI



Outline

1. Introduction
2. Data Description
3. Methodology
4. Results
5. Implications and Limitations

5. Conclusion - Implications

Research Question:

What is most important to consumers?

1. Price (46%)
2. Headquarter Location (27%)
3. Ecological Sustainability (10%)

What customer characteristic influence choice?

1. Age
2. Monthly Clothing Spending
3. Buys Second-hand Clothing

5. Conclusion - Limitations

Model Limitations:

Focus is on store choice,
individual heterogeneity only
captured through the ASC

Fix ☐ Mixed Logit

- Estimate SD for each store factor

or

- Individual Random Effects

However,
limited by 6 choice tasks per person

Data & External Validity:

MLU students:
young, educated, German

→ Limits Generalisability

Hypothetical Behaviour
≠ Buying Behaviour

Product held constant (Black T-Shirt)

Thank You for Listening!

Questions?

References

Carrington, M.J., Neville, B.A. & Whitwell, G.J. Why Ethical Consumers Don't Walk Their Talk: Towards a Framework for Understanding the Gap Between the Ethical Purchase Intentions and Actual Buying Behaviour of Ethically Minded Consumers. *J Bus Ethics* 97, 139–158 (2010). <https://doi.org/10.1007/s10551-010-0501-6>

Niinimäki, K., Peters, G., Dahlbo, H., Perry, P., Rissanen, T., & Gwilt, A. (2020). The environmental price of fast fashion. *Nature Reviews Earth & Environment*, 1, 189–200. <https://doi.org/10.1038/s43017-020-0039-9>